AUTOSAR ADAPTIVE PLATFORM SPECIFICATION
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1. SYSTEM ARCHITECTURE OVERVIEW
The Adaptive Powertrain Management System is architected in accordance with AUTOSAR Classic and Adaptiv

2. SOFTWARE COMPONENTS (SWC)
- PowertrainManager_SWC:
  The master supervisory controller executing vehicle operating mode management (Park, Drive, Neutral, Revers

- TorqueArbitration_SWC:
  The core arbitration unit responsible for merging torque demands from the accelerator pedal, ADAS Adaptive C

- BatterySafety_SWC:
  High-voltage battery supervisor monitoring pack voltage, cell temperature gradients, and contactor weld detecti

- InverterActuator_SWC:
  Low-level hardware abstraction component interfacing with the dual motor inverter bridges via high-speed CAN

3. RUNNABLES AND TASK SCHEDULING
The software architecture defines deterministic periodic and event-driven execution entities:

- Runnable_ArbitrateTorque:
  Periodic runnable scheduled every 10ms with priority 5. It executes the core PID arbitration algorithm, evaluates

- Runnable_CalculateThermals:
  Periodic runnable scheduled every 50ms with priority 3. It calculates stator and rotor thermal models to predict

- Runnable_EmergencyCutoff:
  Event-triggered runnable executed immediately upon receiving high-voltage interlock loop (HVIL) fault signals o

4. AUTOSAR INTERFACES & PORTS
Communication across components is strictly governed by AUTOSAR Port Interfaces:

- TorqueRequest_IF:
  Client-Server interface providing operations GetDriverDemand() and SetArbitratedTorque(). Allowed torque limi

- BatteryStatus_IF:
  Sender-Receiver interface broadcasting state-of-charge (SoC: 0.0% to 100.0%), bus voltage, and isolation resis

- ContactorCommand_IF:
  Dedicated secure interface triggering pyrofuse detachment and main DC contactor disconnect during unrecove

5. SAFETY INTEGRITY & ENTERPRISE CONSTRAINTS
- ISO 26262 ASIL Rating:
  The complete Powertrain management subsystem is classified as ASIL D. The Fault Tolerant Time Interval (FT

- Memory & Buffer Boundaries:
  All arrays must be statically bounded in compliance with MISRA C:2012 Rule 17.2 and AUTOSAR Memory Map

- Inter-Process Communication (IPC) Rules:
  All data exchanges between PowertrainManager_SWC and TorqueArbitration_SWC must pass through the AU